

Fall 2025 CPTS530 Homework 4
MATH 448-548
Numerical Analysis

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Problem 1

Problem Statement (Textbook 4.1.7):

Let A have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are $n \times n$. Prove that if $(B - I)$ is nonsingular, then, for $k \geq 1$,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

Solution:

To understand the pattern, let's first compute A^2 and A^3 directly.

Computing A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \end{aligned}$$

Computing A^3 :

$$\begin{aligned} A^3 &= A^2 \cdot A = \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^3 & B^2C + BC + C \\ 0 & I \end{bmatrix} \end{aligned}$$

Observing the pattern in the upper-right block:

$$\begin{aligned} k = 1 : & \quad C \\ k = 2 : & \quad BC + C \\ k = 3 : & \quad B^2C + BC + C \end{aligned}$$

We can factor these as:

$$\begin{aligned} k = 1 : & \quad (I)C \\ k = 2 : & \quad (B + I)C \\ k = 3 : & \quad (B^2 + B + I)C \end{aligned}$$

This suggests the general formula: the upper-right block is $(I + B + B^2 + \cdots + B^{k-1})C$, which equals $(B^k - I)(B - I)^{-1}C$ (see Appendix A).

Now we prove this result by induction on k .

Base Case ($k = 1$):

For $k = 1$, we need to verify that

$$A^1 = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} = \begin{bmatrix} B^1 & (B^1 - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

This simplifies to showing that $C = (B - I)(B - I)^{-1}C$, which is trivially true since $(B - I)(B - I)^{-1} = I$.

Inductive Step:

Assume the formula holds for $k = n$:

$$A^n = \begin{bmatrix} B^n & (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

We need to prove it holds for $k = n + 1$. Computing $A^{n+1} = A^n \cdot A$:

$$\begin{aligned} A^{n+1} &= \begin{bmatrix} B^n & (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^n \cdot B + (B^n - I)(B - I)^{-1}C \cdot 0 & B^n \cdot C + (B^n - I)(B - I)^{-1}C \cdot I \\ 0 \cdot B + I \cdot 0 & 0 \cdot C + I \cdot I \end{bmatrix} \\ &= \begin{bmatrix} B^{n+1} & B^n C + (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix} \end{aligned}$$

Now we need to show that the upper-right block equals $(B^{n+1} - I)(B - I)^{-1}C$.

Key Identity:

We claim that $(B^{n+1} - I)(B - I)^{-1}C = B^n C + (B^n - I)(B - I)^{-1}C$.

This identity is equivalent to showing that:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \dots + B^{k-1}$$

The proof of this geometric series identity for matrices is provided in Appendix A.

Thus, we have shown that:

$$A^{n+1} = \begin{bmatrix} B^{n+1} & (B^{n+1} - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

By the principle of mathematical induction, the formula holds for all $k \geq 1$.

Problem 2

Problem Statement (Textbook 4.1.12):

Let A be an $n \times n$ invertible matrix, and let u and v be two vectors in $\mathbb{R}^n$. Find the necessary and sufficient conditions on u and v in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

Solution:

Problem 3

Problem Statement (Textbook 4.1.17):

A square matrix A is said to be **skew-symmetric** if $A^T = -A$. Prove that if A is skew-symmetric, then $x^T Ax = 0$ for all x .

Solution:

Let A be a skew-symmetric matrix, so $A^T = -A$. Consider the scalar quantity $x^T Ax$.

Taking the transpose of this scalar (which equals itself):

$$\begin{aligned}(x^T Ax)^T &= x^T A^T (x^T)^T \\ &= x^T A^T x \\ &= x^T (-A)x \\ &= -x^T Ax\end{aligned}$$

Since $x^T Ax$ is a scalar and equals its own negative, we have $x^T Ax = -x^T Ax$, which implies $2x^T Ax = 0$.

Therefore, $x^T Ax = 0$ for all x . □

Problem 4

Problem Statement (Textbook 4.1.18):

(Continuation) Prove that the diagonal elements of a skew-symmetric matrix are **0**. Also, prove that the determinant is 0 when the matrix is of odd order.

Solution:

Problem 5

Problem Statement (Textbook 4.1.19):

(Continuation) Let A be any square matrix, and define $A_0 = \frac{1}{2}(A + A^T)$ and $A_1 = \frac{1}{2}(A - A^T)$. Prove that A_0 is symmetric, that A_1 is skew-symmetric, that $A = A_0 + A_1$ and that for all x , $x^T A x = x^T A_0 x$. This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

Solution:

Problem 6

Part (a)

Problem Statement (Textbook 4.2.1a):

Prove these facts, needed in the proof of Theorem 2.

(a) If U is upper triangular and invertible, then U^{-1} is upper triangular.

Solution:

Part (b)

Problem Statement (Textbook 4.2.1b):

Prove these facts, needed in the proof of Theorem 2.

(b) The inverse of a unit lower triangular matrix is unit lower triangular.

Solution:

Part (c)

Problem Statement (Textbook 4.2.1c):

Prove these facts, needed in the proof of Theorem 2.

(c) The product of two upper (lower) triangular matrices is upper (lower) triangular.

Solution:

Problem 7

Problem Statement (Textbook 4.2.5):

Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

Solution:

A Matrix Geometric Series

Lemma

For any square matrix B such that $(B - I)$ is nonsingular, and for any $k \geq 1$:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \cdots + B^{k-1}$$

Proof:

Define the partial sum sequence:

$$\begin{aligned} S_0 &= 0 \\ S_1 &= I \\ S_2 &= I + B \\ S_3 &= I + B + B^2 \\ &\vdots \\ S_k &= I + B + B^2 + \cdots + B^{k-1} \end{aligned}$$

We will show that $(B - I)S_k = B^k - I$ for all $k \geq 1$.

Multiplying S_k by $(B - I)$:

$$\begin{aligned} (B - I)S_k &= (B - I)(I + B + B^2 + \cdots + B^{k-1}) \\ &= B(I + B + B^2 + \cdots + B^{k-1}) - (I + B + B^2 + \cdots + B^{k-1}) \\ &= (B + B^2 + B^3 + \cdots + B^k) - (I + B + B^2 + \cdots + B^{k-1}) \\ &= B^k - I \end{aligned}$$

Since $(B - I)$ is nonsingular by assumption, we can multiply both sides by $(B - I)^{-1}$ on the left:

$$S_k = (B^k - I)(B - I)^{-1}$$

Therefore:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \cdots + B^{k-1}$$

This completes the proof. □